Calibrated Forecasting and Match Leverage in the Serie A

A correction and extension of *Soccer Match Prediction in the Serie A*

Yannik Pitcan
pitcany@gmail.com

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Abstract

A 2016 study of mine, accepted for publication in the *International Journal of Computer Applications* in 2018, reported 53.0% three-way accuracy on Serie A match outcomes and compared that favourably against betting-market accuracy of 55.3%. This paper corrects that study and extends it from ten seasons to nineteen (7,220 matches, 2007–08 to 2025–26). Three corrections matter. The benchmark should have been the outcome base rate, 40.6%, not uniform guessing at 33%. Bookmaker odds were used as model *features* and then treated as an independent standard of comparison, which inverts the logic of the test. And with 218 scored matches, the reported spread across five classifiers lay inside two standard errors, so no ranking among them was supported.

Rebuilt on a proper likelihood with tuned time decay and rolling-origin validation, the model reaches 53.4% accuracy—statistically indistinguishable from the 2016 figure—and loses to a margin-free closing price by 0.0067 Ranked Probability Score (95% CI $[0.0046, 0.0088]$). In a logarithmic opinion pool with the market, the optimal weight on the structural model is 0.000, on validation and on test alike. Refitting the same machinery to shots on target, a pre-expected-goals proxy for chance creation, earns weight 0.35 against the goals model but again 0.000 against the market: two signals, each carrying information the other lacks, both already priced.

The constructive consequence is that the deliverable is not a better outcome forecast but what is built on a calibrated one. We define *match leverage*—the change in a club's probability of achieving its season objective between winning and losing a given fixture—and compute it for ACF Fiorentina from 1 March 2026. The ordering inverts intuition: an away fixture against a relegation rival carried 2.3× the leverage of hosting the eventual champions.

1 Introduction

Forecasting association-football outcomes is a mature problem with an uncomfortable property: the most accessible benchmark, the betting market, is also a very strong one. Published work in the applied-computing literature frequently reports accuracy figures in the low fifties and presents them as competitive without establishing what the relevant floor and ceiling are. My own 2016 study (Pitcan, 2018) is an instance of this, and the present paper begins by correcting it.

Section 2 documents the specific errors. Sections 3 and 4 describe the corpus and the rebuilt methodology. Section 5 reports the central negative result and the shots-on-target replication. Section 6 defines match leverage and applies it. Section 7 states what the work cannot do.

Two framing errors drove the original conclusion, and both are worth stating before the arithmetic. First, the benchmark: the study compared its accuracy against uniform guessing at 33%. Home advantage alone places the majority class well above that—over this corpus the home side wins 44.1% of matches—so roughly seven percentage points of the apparent margin were never a margin at all. Second, and more seriously, bookmaker odds were supplied to the classifiers as input features, after which the resulting accuracy was compared against the accuracy of the bookmaker. A model given the market's own forecast should match it as a floor; scoring below it indicates the pipeline is discarding information rather than adding any. The original noted the symptom ("the betting odds did not impact predictions very much") and attributed it to the odds being uninformative.

2 Errata to the 2016 study

The following are recomputed directly from the confusion matrices printed in Pitcan (2018). Both matrices total 218 matches, against a stated test set of 320; the difference is early-season fixtures dropped for insufficient match history, a discrepancy the text does not reconcile.

A mislabelled table. Tables 4.1 and 4.2 of the original are both captioned "Linear SVM". Table 4.1 has diagonal $80 + 29 + 2 = 111$, i.e. 50.9%, matching the reported Linear SVM figure. Table 4.2 has diagonal $77 + 34 + 3 = 114$, i.e. 52.3%, which is the reported AdaBoost figure. Table 4.2 is AdaBoost.

Binary classifiers below the trivial baseline. Away wins constitute $60/218 = 27.5\%$ of the test set, so the constant rule "never predict an away win" scores 72.5%. The reported away-win accuracies are 72.9 (Gaussian SVM), 69.7 (Linear SVM), 68.4 (AdaBoost) and 45.9 (logistic regression). None materially exceeds the constant rule; three fall below it. On the home side, home wins are 92/218, so "never predict a home win" scores $126/218 = 57.80\%$. The reported Linear SVM home accuracy is 57.8. To three significant figures this is the constant rule, which implies the classifier predicted no home win at all.

Model ranking within noise. With $n = 218$, the standard error of an accuracy near one half is $\sqrt{0.25/218} \approx 0.034$. The reported multi-class accuracies span 47.2% to 53.0%—under two standard errors end to end—and the $n = 5$ versus $n = 7$ comparison moves the three classifiers in inconsistent directions. The conclusion that a particular classifier performed best is not supported.

An error in the form weighting. The original defines a time-weighted form feature as $\tau'(\{x_1, \ldots, x_n\}) = \frac{1}{n} \sum_i x_i e^{-i}$. A weighted average must be normalised by $\sum_i w_i$, not by n. With $n = 5$, $\sum_{i=1}^{5} e^{-i} = 0.578$, so dividing by 5 shrinks the weighted feature to roughly 1/8.7 of the scale of its unweighted counterpart—material for the kernel and regularised methods used, which are not scale-invariant. Separately, the decay is severe: 63.6% of the weight falls on the most recent match and the Kish effective sample size is 2.1 matches, so a nominal five-match window carries about two matches of information. The decay rate was fixed by assertion rather than fitted.

3 Data

The corpus is nineteen complete Serie A seasons, 2007–08 through 2025–26, comprising 7,220 matches, obtained from `football-data.co.uk`. Each record carries date, clubs, full-time score and outcome, shots, shots on target and corners for both sides, and closing-line prices. Bet365 opening prices are present for all seasons; Pinnacle opening and closing prices are present from 2012–13, and the closing line is preferred wherever available.

One structural feature of the corpus deserves note, because it motivates a modelling choice below: promotion and relegation mean only eight clubs appear in all nineteen seasons, while many appear once or twice. Any rating scheme must therefore handle clubs with little or no top-flight history.

A second feature is substantive rather than technical. Home advantage in Serie A has eroded materially over the period: the home win rate falls from 47.5% (2007–13) to 40.2% (2023–26), and the ratio of home to away goals from 1.38 to 1.14. A model fitted on the full corpus without time weighting would misstate the current strength of home advantage.

4 Methods

4.1 Market probabilities

Decimal odds are a reciprocal transform of probability, and the reciprocals of a quoted book sum to more than one by the bookmaker's margin. Rather than supply raw odds as features, we convert them to probabilities and remove the margin by Shin's insider-trading model (Shin, 1993), which apportions more of the margin to longshots than proportional normalisation does and is the best-performing practical method in the comparison of Štrumbelj (2014). For a k-outcome book with raw inverse odds π_i and book sum $\Pi = \sum_j \pi_j$, the implied probabilities solve

$$p_i(z) = \frac{\sqrt{z^2 + 4(1-z)\,\pi_i^2/\Pi} - z}{2(1-z)}, \qquad \text{with } z \text{ chosen so } \sum_i p_i(z) = 1. \tag{1}$$

The row sum is strictly decreasing in z, so the root is located by bisection.

4.2 Structural model

Scorelines are modelled with the bivariate Poisson construction of Dixon and Coles (1997). Each club j carries an attack rating α_j and a defence rating δ_j; for a fixture between home club h and away club a,

$$\log\lambda = \gamma + \alpha_h - \delta_a, \qquad \log\mu = \alpha_a - \delta_h, \tag{2}$$

with γ the home advantage. The joint mass on low scores is corrected by the factor $\tau(x, y)$ of Dixon and Coles (1997), which adjusts the four cells $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$ through a dependence parameter ρ; the independent-Poisson product demonstrably misfits there. Matches are weighted exponentially by age,

$$w(t) = \exp(-\xi\,\Delta t)\,, \tag{3}$$

with Δt in days before the forecast origin. Attack ratings are constrained to sum to zero, which removes the one-dimensional translation invariance of the pair (α, δ) and renders the parameters identifiable. Clubs absent from the training window—newly promoted sides—take the exposure-weighted league mean rather than an arbitrary constant.

The decay rate ξ is selected on the validation period rather than assumed. The grid search returns a clean interior optimum at $\xi = 0.002$ per day, a half-life of 347 days, appreciably slower than the 0.0065 of the original Dixon–Coles paper.

4.3 A process-based variant

Goals are a sparse realisation of chance creation, and a rating fitted to them inherits finishing noise. Expected goals is the standard remedy; no freely licensed source covers Serie A back to 2007. We therefore fit the identical machinery to *shots on target*, the pre-expected-goals proxy the corpus already contains, and convert the predicted shot rates to goal rates with a single league-wide finishing factor estimated on the same window. The factor is deliberately not club-specific, since club finishing variation is the noise the substitution is intended to remove.

4.4 Evaluation

Because the outcome is ordered—home win, draw, away win—the primary metric is the Ranked Probability Score (Epstein, 1969), which is the standard choice in football forecasting following Constantinou and Fenton (2012):

$$\mathrm{RPS} = \frac{1}{r-1}\sum_{i=1}^{r-1}\left(\sum_{j=1}^{i}(p_j - a_j)\right)^2, \tag{4}$$

with $r = 3$. RPS penalises a confident home call more heavily when the away side wins than when the match is drawn; accuracy and the Brier score cannot make that distinction. Log loss and Brier are reported alongside, and accuracy is retained solely for comparability with the 2016 study.

All models are scored walk-forward: fitted on matches strictly preceding an origin, used to forecast the following fortnight, then refitted. The original used five-fold cross-validation on time-ordered data, which permits a fold containing March to inform a prediction about January. Hyperparameters, the pooling weight and the calibration temperature are fixed on 2013–14 to 2018–19 (n = 2,280); the test period, 2019–20 to 2025–26 (n = 2,660), is scored once. Every reported mean carries a bootstrap interval, and model comparisons use a paired bootstrap over matches, exploiting the fact that competing models score the same fixtures.

4.5 The incremental-information test

The question the 2016 study should have posed is not whether the model is accurate but whether it carries information the market has not already priced. A logarithmic opinion pool answers this directly. For market forecast p^M and model forecast p^S,

$$p \;\propto\; \left(p^M\right)^{1-w}\left(p^S\right)^{w}, \tag{5}$$

with w fitted out of sample. A weight indistinguishable from zero is evidence that the model adds nothing; a weight materially above zero is evidence of genuine incremental information. The logarithmic pool is preferred to a linear one as it is externally Bayesian and operates naturally on log-odds.

Table 1: Test period, 2019–20 to 2025–26 ($n = 2{,}660$). Lower RPS is better.

Model	RPS	95% CI	Log loss	Brier	Accuracy
Market (Shin de-vigged)	**0.1905**	[0.1856, 0.1957]	0.9620	0.5717	54.8%
Market + model pool	0.1905	[0.1856, 0.1957]	0.9620	0.5717	54.8%
Dixon–Coles (calibrated)	0.1972	[0.1921, 0.2024]	0.9858	0.5862	53.4%
Shots on target	0.2001	[0.1959, 0.2042]	0.9959	—	52.9%
Base rate	0.2318	[0.2292, 0.2344]	1.0846	0.6574	40.6%
Uniform (2016 benchmark)	0.2340	[0.2312, 0.2367]	1.0986	0.6667	40.6%

Table 2: Pool weights fitted on the validation period.

Pool	Market	Goals model	Shots model
Market + shots	1.00	—	0.00
Goals + shots	—	0.65	**0.35**
Market + goals + shots	1.00	0.00	0.00

5 Results

Table 1 reports the central comparison. The market attains RPS 0.1905; the structural model 0.1972. The paired bootstrap places the difference at +0.0067 RPS with 95% interval $[0.0046, 0.0088]$—small in magnitude but decisively non-zero. The model is worse than the market, not comparable to it.

Two secondary observations follow. First, the rebuilt model attains 53.4% accuracy against the 2016 study's 53.0%. Nine additional seasons, a principled likelihood, tuned decay and leak-free validation bought 0.4 accuracy points, well inside noise. This is informative rather than disappointing: three-way match accuracy appears to be near its ceiling, and effort spent improving it is spent on a quantity that has stopped moving. Second, the correct floor is the base rate at 40.6%, not uniform guessing.

The pooling weight. Fitted on validation, the optimal weight on the structural model in a logarithmic pool with the market is 0.000. This is not a small weight but a boundary solution: log loss increases monotonically as any weight is transferred onto the model, on the validation period and—were one to cheat and fit it there—on the test period also.

Widening the scan to negative weights, which fall outside the admissible range for a pool, the validation minimum is interior at $\hat{w} = -0.225$; carried to the test period it improves log loss by 0.00062 (95% CI $[-0.00105, 0.00231]$) and RPS by 0.00021 (95% CI $[-0.00029, 0.00071]$), both intervals containing zero. A negative weight divides the market forecast by the flatter structural one and so sharpens it, which is a temperature correction on the price rather than a contribution of information.

Is the target simply too noisy? The natural objection is that the model failed because goals are noisy, not because the method is inadequate. Table 2 addresses this. Against the goals model alone, the shots-on-target variant earns weight 0.35: it demonstrably carries information the goals model lacks, so the negative result is not an artefact of target choice. Against the market, the same signal earns 0.00, and in the three-way pool both structural models collapse to zero simultaneously. Two model families built on different targets, each holding information the other lacks, and the

Table 3: ACF Fiorentina, simulated from 1 March 2026. The club stood 16th on 24 points with twelve fixtures remaining. Objective: survival, i.e. a final position of 17th or better.

Fixture	Venue	$\Pr(\Omega \mid W)$	$\Pr(\Omega \mid D)$	$\Pr(\Omega \mid L)$	Leverage
Lecce	A	0.987	0.959	0.914	**0.073**
Cremonese	A	0.986	0.957	0.922	0.063
Verona	A	0.982	0.952	0.927	0.055
Genoa	H	0.984	0.962	0.932	0.053
Udinese	A	0.985	0.961	0.933	0.052
Sassuolo	H	0.982	0.956	0.932	0.050
Lazio	H	0.988	0.960	0.942	0.046
Parma	H	0.981	0.955	0.936	0.045
Atalanta	H	0.990	0.966	0.949	0.041
Juventus	A	0.988	0.971	0.951	0.037
Roma	A	0.988	0.968	0.952	0.036
Inter	H	0.988	0.971	0.955	**0.033**

closing price has already absorbed both.

Calibration and discrimination are distinct. On the home-win margin the structural model is *better calibrated* than the market—calibration slope 0.995 against the market's 1.103, with comparable expected calibration error (0.021 for both)—while being clearly less sharp. The fitted recalibration temperature is 0.99, i.e. essentially no correction is warranted. The market's advantage is therefore information, not honesty. These are different deficiencies and they call for different remedies; a study reporting only accuracy cannot distinguish them.

6 Match leverage

If a structural model cannot beat a closing price on match outcomes, the product worth building is not a match-outcome forecast. A club can read the market for free. What a club cannot read off the market is the quantity that governs its decisions: how much a particular fixture matters.

Let Ω denote a season objective—survival, European qualification, the title—expressed as a band of final league positions. For a remaining fixture m, define

$$L_m \;=\; \Pr(\Omega \mid \text{win } m) \;-\; \Pr(\Omega \mid \text{lose } m)\,. \tag{6}$$

We estimate L_m by simulating the remainder of the season 20,000 times, sampling complete scorelines from each fixture's joint distribution so that goal difference resolves naturally, and conditioning on the simulated result of m. Because fixtures are sampled independently, conditioning after the fact is equivalent to intervening on m and resampling the remainder; this equivalence was verified by exhaustive enumeration on a reduced league and by direct comparison against forced-outcome simulation at 10^6 replications.

Table 3 reports the result. The ordering inverts intuition. The fixtures that move survival most are those against direct relegation rivals; those against the leading clubs move it least, because the model already expects defeat in both branches and has priced it. Losing at home to Inter—the eventual champions, and by any measure the season's marquee fixture—costs Fiorentina 3.3 percentage points of survival probability. Losing away at Lecce costs 7.3. The away fixture carries

2.3× the leverage of the home one, and the gap is approximately eight Monte Carlo standard errors wide, so the ordering is not simulation noise.

This is a rotation and load-management argument expressed in probability. The Inter fixture is where a fatigued first-choice player is cheapest to rest; Lecce away is where he is most expensive. No match-outcome forecast conveys that on its own.

As a coherence check rather than a claim of predictive success, the simulation projected 41.4 expected final points against an actual 42, and assigned a survival probability of 96.4%; Fiorentina finished 15th. A single realisation from a distribution is weak evidence, but it establishes that the season aggregation is not obviously broken, which is the minimum bar before a leverage ranking is acted upon.

7 Limitations

No player-level information. The model has no access to injuries, suspensions, rest days, or transfers. This is the largest gap between this work and what a club holds internally, and the most likely location of a genuine edge over a market price.

No expected goals. Shots on target weight a tap-in and a speculative long-range effort equally. Licensed event data would sharpen the process-based variant, though the results in Table 2 suggest it would refine the conclusion rather than reverse it.

The benchmark is a closing price. It incorporates late team news the model never observes. Some portion of the 0.0067 RPS gap is information asymmetry rather than modelling deficiency.

Leverage is single-objective. It conditions on one position band at a time. A club trading survival against a cup run requires a utility function, not a probability difference.

Tiebreakers are simplified to points, then goal difference, then goals scored; Serie A resolves equal points on head-to-head record first. Exact ties affecting the reported band occur in under 0.01% of simulations.

8 Reproducibility

All results derive from public data and a versioned Python package. The implementation carries 126 tests. Four areas—the margin-removal solver, the likelihood and its identifiability, the walk-forward harness, and the leverage estimator—were additionally subjected to independent adversarial audit, which confirmed the absence of look-ahead bias by positive control and identified four defects, all corrected before the results reported here were generated. Two of the corrected defects were tests that asserted nothing: one passed against a deliberately leaky harness, the other checked only that probabilities summed to one. Code and data pipeline: `https://github.com/pitcany/seriea-leverage`.

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